

Project Setup & Imports

English

We add the project root to `sys.path` so the notebook can import directly from `src/`, exactly like the scripts in `src/` import from each other. This works regardless of whether the notebook is launched from `notebooks/` or from the project root.

العربية

`src/` حتى يتمكن الدفتر من الاستيراد مباشرة من مجلد `sys.path` نقوم بإضافة جذر المشروع إلى `src/` تمامًا كما تستورد الملفات داخل من بعضها البعض. يعمل هذا بغض النظر عما إذا تم تشغيل `src/` أو من جذر المشروع `notebooks/` الدفتر من

```
In [53]: import sys
from pathlib import Path

import pandas as pd
from IPython.display import display, Image

# -----
# Make src/ importable, regardless of where the notebook is launched from
# -----
NOTEBOOK_DIR = Path.cwd()
PROJECT_ROOT = NOTEBOOK_DIR.parent if NOTEBOOK_DIR.name == "notebooks" else NOTEBOOK_DIR

if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))

# -----
# Project modules
# -----
from src.data_loader import (
    load_raw_data,
    load_processed_data,
    get_dataset_info,
    PROCESSED_DATA_FILE,
)

from src.data_cleaning import run_cleaning_pipeline

from src.data_analysis import (
    generate_analysis_report,
    analyze_target,
    get_correlation,
)

from src.visualization import (
    FIGURES_DIR,
    plot_histogram,
    plot_boxplot,
    plot_scatter,
    plot_regression,
    plot_correlation_heatmap,
    plot_bar_chart,
```

```

plot_line,
plot_pairplot,
plot_countplot,
plot_calories_distribution,
plot_calories_vs_time,
plot_calories_vs_distance,
plot_calories_vs_heart_rate,
plot_calories_vs_weight,
plot_violin,
plot_kde,
plot_target_correlations,
plot_calories_by_gender,
plot_scatter_by_gender,
)

pd.set_option("display.max_columns", None)

print(f"Project root : {PROJECT_ROOT}")
print(f"Figures dir : {FIGURES_DIR}")

```

Project root : c:\Users\ayman\OneDrive\Desktop\calories-prediction

Figures dir : C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures

Display Helper

English: `src/visualization.py` already saves every figure to `reports/figures/` and closes it (so the pipeline can run headlessly). To still see each figure inline in this notebook, we add one small helper that shows whichever PNG(s) were just created, right after calling the plotting function. No plotting logic lives here — it only displays what `src/visualization.py` already saved.

وإغلاقه `reports/figures/` بالفعل بحفظ كل رسم في `src/visualization.py` العربي: تقوم (حتى يعمل خط الأنابيب دون واجهة رسومية). لعرض كل رسم داخل هذا الدفتر مباشرة، نضيف دالة التي تم إنشاؤها للتو، مباشرة بعد استدعاء دالة الرسم. لا PNG مساعدة صغيرة تعرض ملف/ملفات بحفظه بالفعل `src/visualization.py` يوجد أي منطق رسم هنا — فهي تعرض فقط ما قامت

```

In [54]: def call_and_show(func, *args, **kwargs):
          """
          Call a plotting function from src.visualization, then display
          whichever PNG file(s) it just saved into FIGURES_DIR.
          """

          before = {f: f.stat().st_mtime for f in FIGURES_DIR.glob("*.png")}

          func(*args, **kwargs)

          after = {f: f.stat().st_mtime for f in FIGURES_DIR.glob("*.png")}

          new_or_updated = [
              f for f, mtime in after.items()
              if f not in before or before[f] != mtime
          ]

```

```
for f in sorted(new_or_updated):
    print(f"Figure saved: {f}")
    display(Image(filename=str(f)))
```

Load Raw Dataset

English

We load the raw dataset using `load_raw_data()` from `src/data_loader.py`, which resolves `data/raw/dataset.csv` relative to the project root itself (not the current working directory), so it works no matter where the notebook is run from.

العربية

والتي ، `src/data_loader.py` من `load_raw_data()` نقوم بتحميل البيانات الخام باستخدام `data/raw/dataset.csv` تحدد مسار (وليس مجلد العمل الحالي)، لذلك تعمل بغض النظر عن مكان تشغيل الدفتر.

```
In [55]: df_raw = load_raw_data()

info = get_dataset_info(df_raw)

for key, value in info.items():
    if key != "column_names":
        print(f"{key}: {value}")

print("\nColumns:")
for column in info["column_names"]:
    print(f"- {column}")

df_raw.head()
```

```
rows: 200
columns: 10
total_values: 2000
missing_values: 0
duplicate_rows: 0
```

```
Columns:
- Gender
- Age
- Height(cm)
- Weight(kg)
- BMI
- Running Time(min)
- Running Speed(km/h)
- Distance(km)
- Average Heart Rate
- Calories Burned
```

Out[55]:

	Gender	Age	Height(cm)	Weight(kg)	BMI	Running Time(min)	Running Speed(km/h)	Distance(km)
0	Male	49	184	74	20.4	78	9.5	3.00
1	Female	56	182	70	23.2	98	10.2	5.23
2	Male	49	170	85	27.4	91	10.4	7.61
3	Male	21	181	59	24.0	89	13.4	14.19
4	Male	47	172	86	29.8	79	9.8	3.84

Exploratory Analysis (Before Cleaning)

English

`generate_analysis_report()` from `src/data_analysis.py` runs every check in one call: basic info, missing values, duplicates, numerical/categorical columns, full statistics, unique-value counts, categorical summary, correlation matrix, and an IQR-based outlier count per column. We inspect the parts most relevant before cleaning: basic info, missing values, duplicates, and outliers.

العربية

تقوم `generate_analysis_report()` من `src/data_analysis.py` بتنفيذ جميع الفحوصات في استدعاء واحد: المعلومات الأساسية، القيم المفقودة، التكرارات، الأعمدة الرقمية/الفئوية، الإحصاءات الكاملة، عدد القيم الفريدة، الملخص الفئوي، مصفوفة الارتباط، وعدد القيم المتطرفة لكل عمود بطريقة نفحص هنا الأجزاء الأكثر أهمية قبل التنظيف: المعلومات الأساسية، القيم المفقودة، التكرارات، IQR، والقيم المتطرفة.

```
In [56]: raw_report = generate_analysis_report(df_raw)

print("Basic info:")
for key, value in raw_report["basic_info"].items():
    print(f" {key}: {value}")

print("\nMissing values (before cleaning):")
print(raw_report["missing_values"])

print(f"\nDuplicate rows (before cleaning): {raw_report['duplicates']}")
```

Basic info:
rows: 200
columns: 10
total_values: 2000
duplicate_rows: 0
missing_values: 0

Missing values (before cleaning):
Series([], dtype: int64)

Duplicate rows (before cleaning): 0

```
In [57]: print("Outlier counts per numerical column (IQR method, before cleaning):")
raw_report["outliers"]
```

Outlier counts per numerical column (IQR method, before cleaning):

Out[57]: Age 0
Height(cm) 0
Weight(kg) 0
BMI 0
Running Time(min) 0
Running Speed(km/h) 0
Distance(km) 0
Average Heart Rate 0
Calories Burned 0
dtype: int64

```
In [58]: raw_report["statistics"]
```

Out[58]:

	count	mean	median	mode	min	max	range	std	var
Age	200	41.38500	44.000	50.00	18.0	64.00	46.00	13.225785	174.9
Height(cm)	200	175.09500	175.000	175.00	150.0	199.00	49.00	14.266170	203.5
Weight(kg)	200	75.18500	74.500	77.00	50.0	99.00	49.00	14.106585	198.9
BMI	200	24.24400	24.600	19.50	18.5	30.00	11.50	3.243558	10.5
Running Time(min)	200	74.34000	74.500	58.00	30.0	119.00	89.00	25.028916	626.4
Running Speed(km/h)	200	11.34150	11.350	11.20	7.0	15.00	8.00	2.167769	4.6
Distance(km)	200	8.92655	8.825	13.23	3.0	14.96	11.96	3.819304	14.5
Average Heart Rate	200	149.82000	150.500	139.00	121.0	179.00	58.00	16.949158	287.2
Calories Burned	200	584.08500	554.000	200.00	200.0	996.00	796.00	228.289659	52116.1



Data Cleaning

English

`run_cleaning_pipeline()` from `src/data_cleaning.py` performs the full sequence — standardize column names, convert numerical columns, clean `Gender` values, fill missing values (median/mode), drop duplicates, and remove invalid/impossible values — then saves the result to `data/processed/dataset_clean.csv` (via `PROCESSED_DATA_FILE`, imported from `src/data_loader.py`) and returns a before/after cleaning report.

العربية

بتنفيذ التسلسل الكامل — `run_cleaning_pipeline()` من `src/data_cleaning.py` تقوم تعبئة القيم المفقودة (بالوسيط)، `Gender`، توحيد أسماء الأعمدة، تحويل الأعمدة الرقمية، تنظيف قيم المنوال، حذف التكرارات، وإزالة القيم غير الصالحة/المستحيلة — ثم تحفظ النتيجة في `data/processed/dataset_clean.csv` (عبر `PROCESSED_DATA_FILE` المستوردة من `src/data_loader.py`) وتُرجع تقريرًا لما قبل وبعد التنظيف.

```
In [59]: cleaned_df, cleaning_report = run_cleaning_pipeline(
          df_raw,
          output_path=PROCESSED_DATA_FILE,
          )

for key, value in cleaning_report.items():
    print(f"{key}: {value}")
```

```
Cleaned dataset saved to: C:\Users\ayman\OneDrive\Desktop\calories-prediction\data\p
rocessed\dataset_clean.csv
original_rows: 200
cleaned_rows: 200
rows_removed: 0
original_columns: 10
cleaned_columns: 10
missing_values_before: 0
missing_values_after: 0
duplicates_before: 0
duplicates_after: 0
```

Load Processed Dataset

English

We reload the saved file with `load_processed_data()` rather than reusing `cleaned_df` in memory, to confirm the file on disk (the one every later stage of the project will actually read) matches what the pipeline produced.

العربية

بدلاً من الاعتماد على `load_processed_data()` نقوم بإعادة تحميل الملف المحفوظ باستخدام `cleaned_df` في الذاكرة، للتأكد من أن الملف الموجود على القرص (وهو ما ستقرأه فعليًا كل مرحلة

لاحقة في المشروع) يطابق ما أنتجه خط الأنايب

```
In [60]: df = load_processed_data()

print(f"Shape: {df.shape}")
df.head()
```

Shape: (200, 10)

Out[60]:

	Gender	Age	Height(cm)	Weight(kg)	BMI	Running Time(min)	Running Speed(km/h)	Distance(km)
0	Male	49	184	74	20.4	78	9.5	3.00
1	Female	56	182	70	23.2	98	10.2	5.23
2	Male	49	170	85	27.4	91	10.4	7.61
3	Male	21	181	59	24.0	89	13.4	14.19
4	Male	47	172	86	29.8	79	9.8	3.84

Post-Cleaning Analysis

English

We re-run the target analysis and correlation matrix on the cleaned data using `analyze_target()` and `get_correlation()` from `src/data_analysis.py`.

العربية

نعيد تشغيل تحليل المتغير المستهدف ومصفوفة الارتباط على البيانات النظيفة باستخدام `analyze_target()` و `get_correlation()` من `src/data_analysis.py`.

```
In [61]: analyze_target(df, "Calories Burned")
```

```
Out[61]: count      200.000000
mean      584.085000
median    554.000000
mode      200.000000
min       200.000000
max       996.000000
range     796.000000
std       228.289659
variance  52116.168618
Q1        395.750000
Q2        554.000000
Q3        791.500000
IQR        395.750000
dtype: float64
```

```
In [62]: correlation_matrix = get_correlation(df)
correlation_matrix
```

Out[62]:

	Age	Height(cm)	Weight(kg)	BMI	Running Time(min)	Running Speed(km/h)	Dist
Age	1.000000	-0.115622	-0.062736	-0.036499	0.008984	-0.113733	
Height(cm)	-0.115622	1.000000	-0.030276	-0.115931	0.112313	-0.028401	
Weight(kg)	-0.062736	-0.030276	1.000000	-0.050567	-0.109841	0.061732	
BMI	-0.036499	-0.115931	-0.050567	1.000000	-0.039732	-0.106841	
Running Time(min)	0.008984	0.112313	-0.109841	-0.039732	1.000000	-0.061102	
Running Speed(km/h)	-0.113733	-0.028401	0.061732	-0.106841	-0.061102	1.000000	
Distance(km)	-0.020051	0.067200	0.018295	0.144888	-0.056769	0.008205	
Average Heart Rate	-0.115899	0.062085	0.017206	-0.046920	-0.035617	-0.010710	
Calories Burned	0.035348	0.100808	0.070512	-0.059128	-0.056699	0.013207	



Visualizations

English

Every chart below is produced entirely by `src/visualization.py` ; this notebook just calls each function (through `call_and_show`) and displays the saved PNG. Numerical features are handled with a loop so adding a new feature to the project does not require adding a new cell here.

العربية

؛ هذا الدفتر فقط يستدعي كل `src/visualization.py` كل رسم أدناه يتم إنتاجه بالكامل بواسطة المحفوظ. تتم معالجة الخصائص الرقمية عبر حلقة PNG ويعرض ملف (`call_and_show` عبر) دالة تكرار حتى لا تتطلب إضافة خاصية جديدة للمشروع إضافة خلية جديدة هنا.

```
In [63]: numerical_features = [
    "Age",
    "Height(cm)",
    "Weight(kg)",
    "BMI",
    "Running Time(min)",
    "Running Speed(km/h)",
```

```
"Distance(km)",  
"Average Heart Rate",  
"Calories Burned",  
]
```

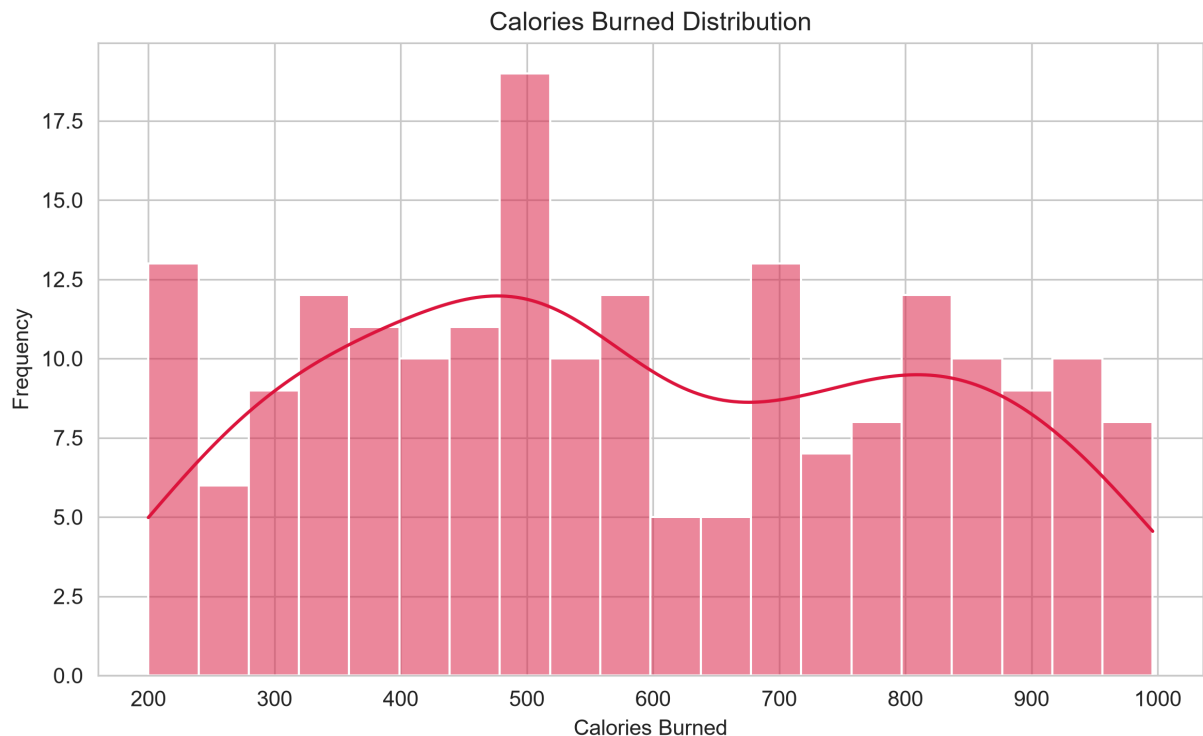
Target Distribution — Calories Burned

```
plot_calories_distribution()
```

```
In [64]: call_and_show(plot_calories_distribution, df)
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\calories_burned_distribution.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\calories_burned_distribution.png



Numerical Feature Distributions — `plot_histogram()`

```
In [65]: for feature in numerical_features:  
         call_and_show(plot_histogram, df, feature)
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Age.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Age.png

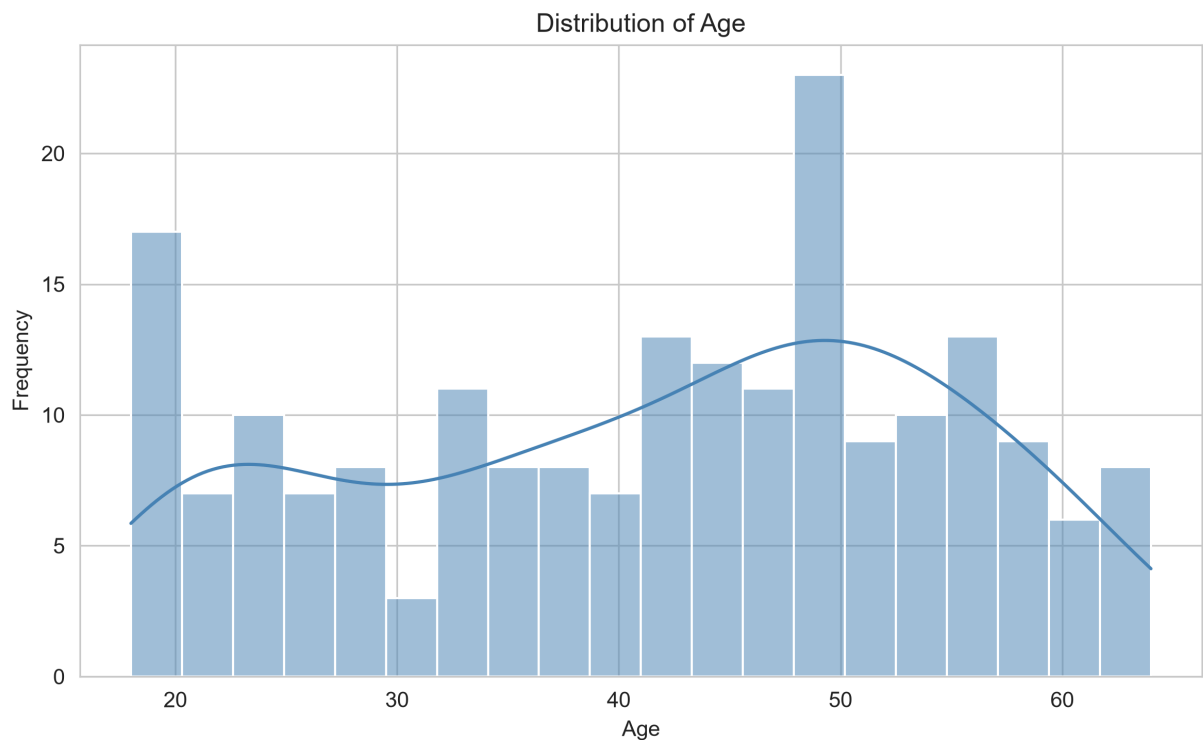


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Height(cm).png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Height(cm).png

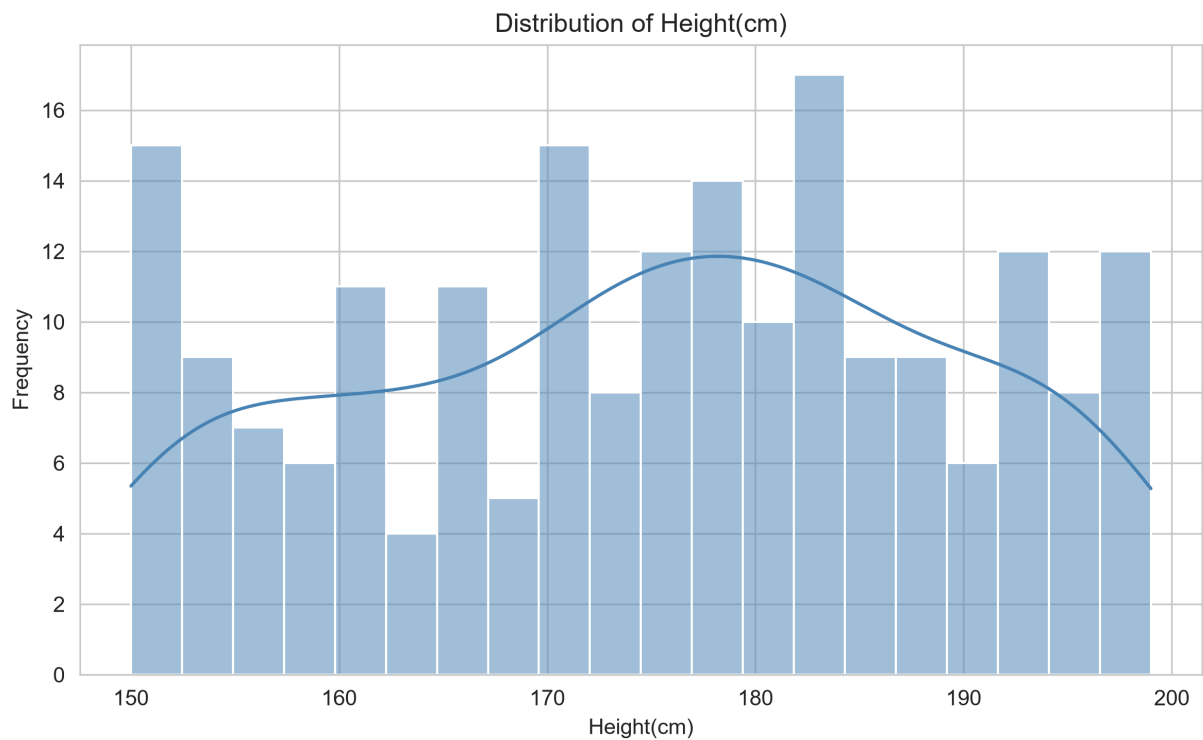


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Weight(kg).png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Weight(kg).png

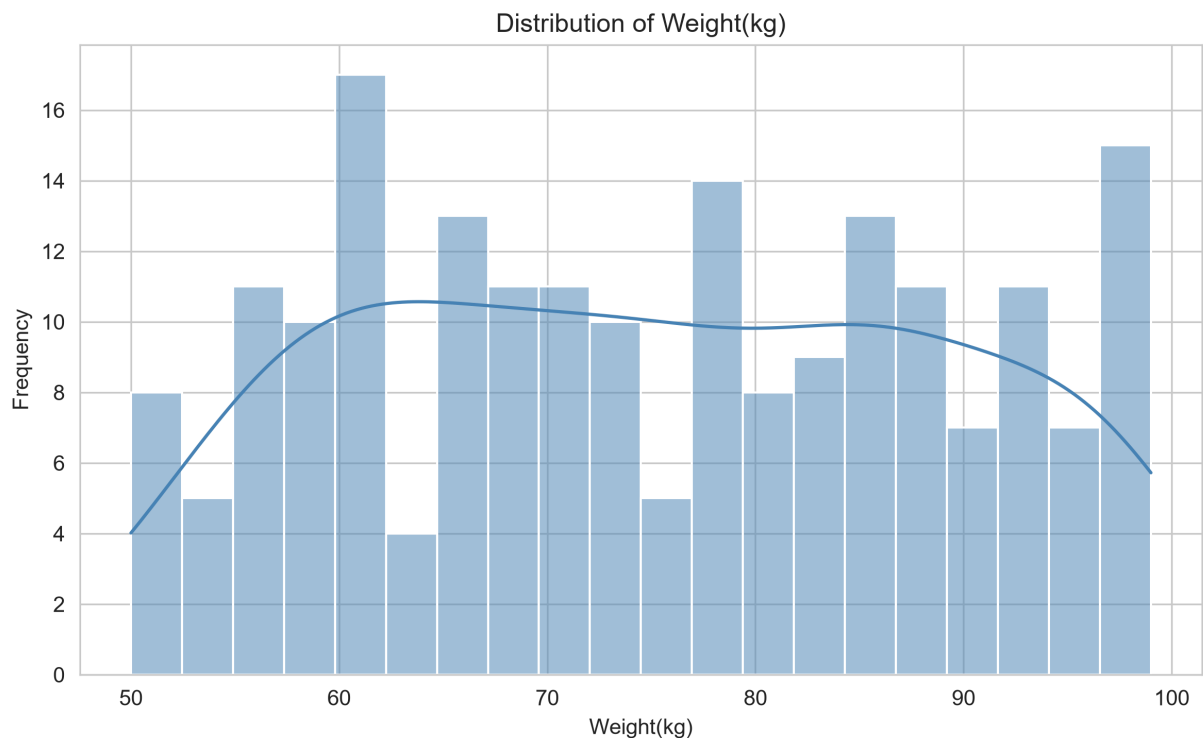


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_BMI.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_BMI.png

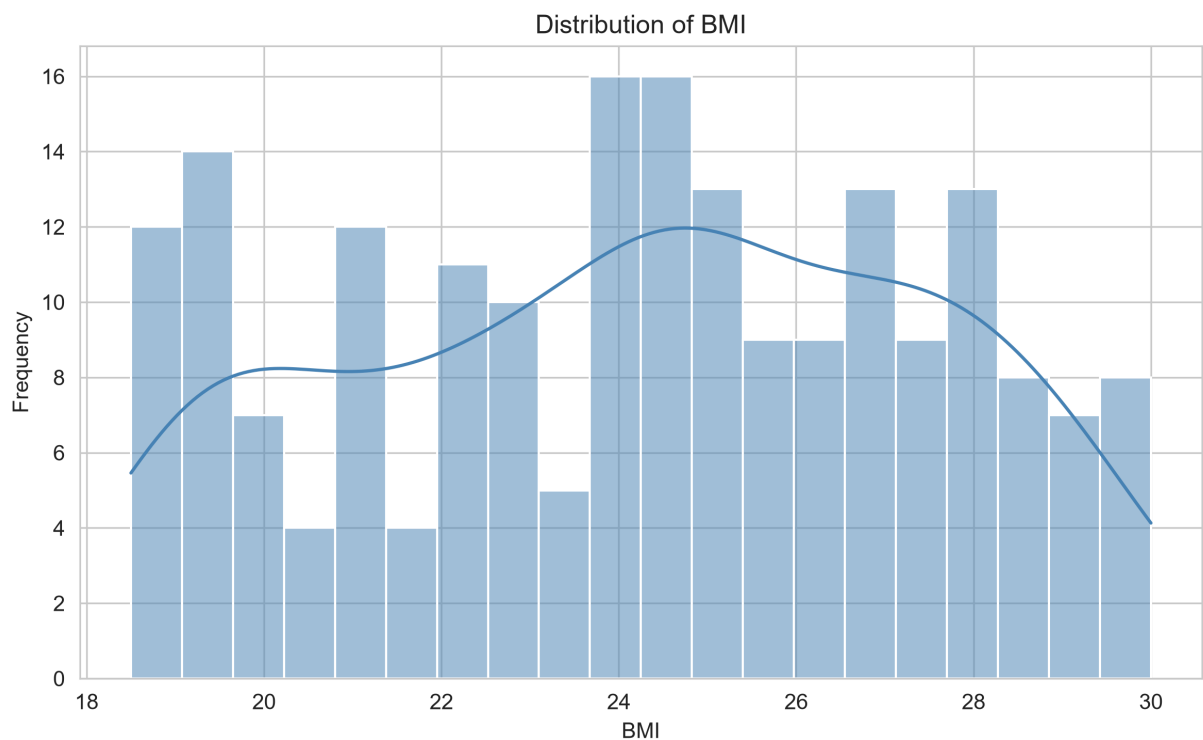


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Running_Time(min).png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Running_Time(min).png

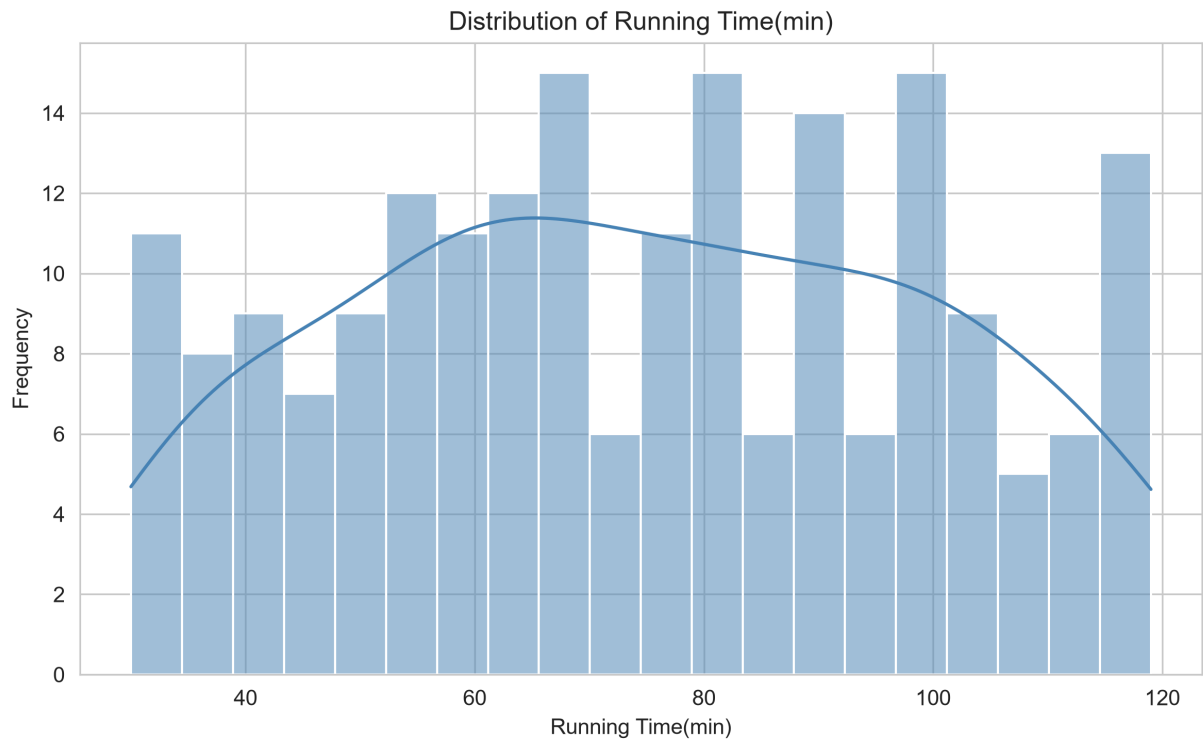


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Running_Speed(km_h).png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Running_Speed(km_h).png

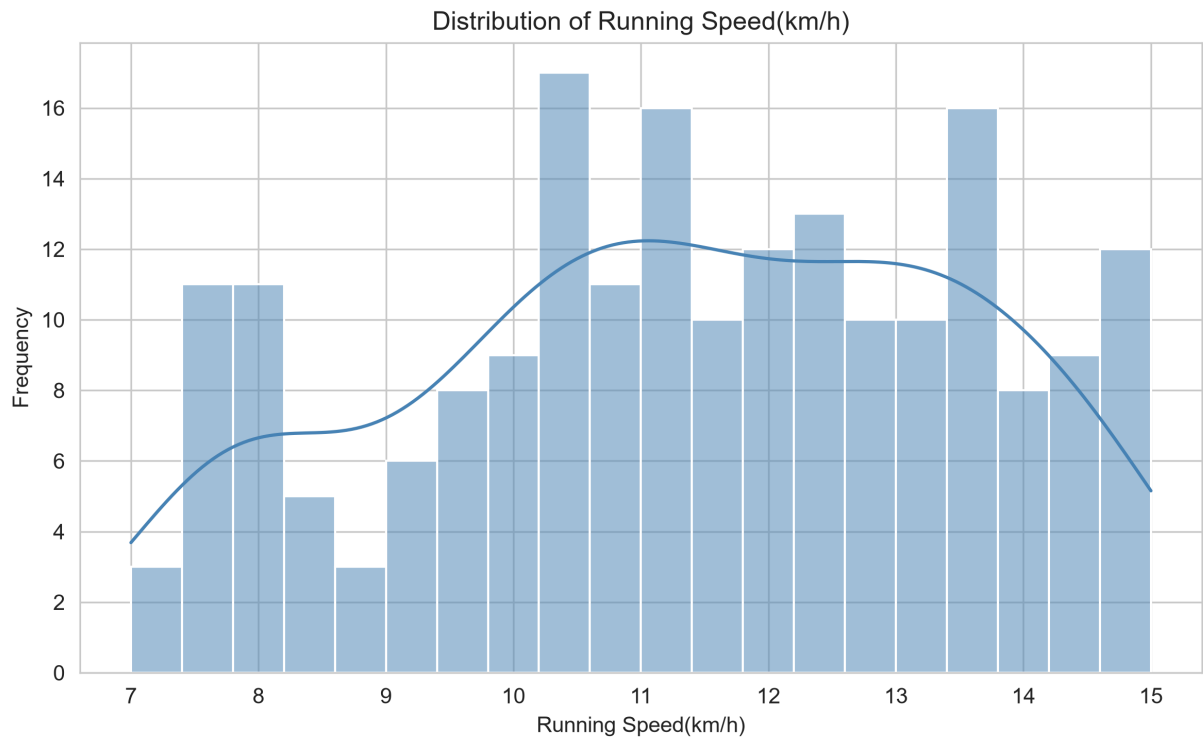


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Distance(km).png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Distance(km).png

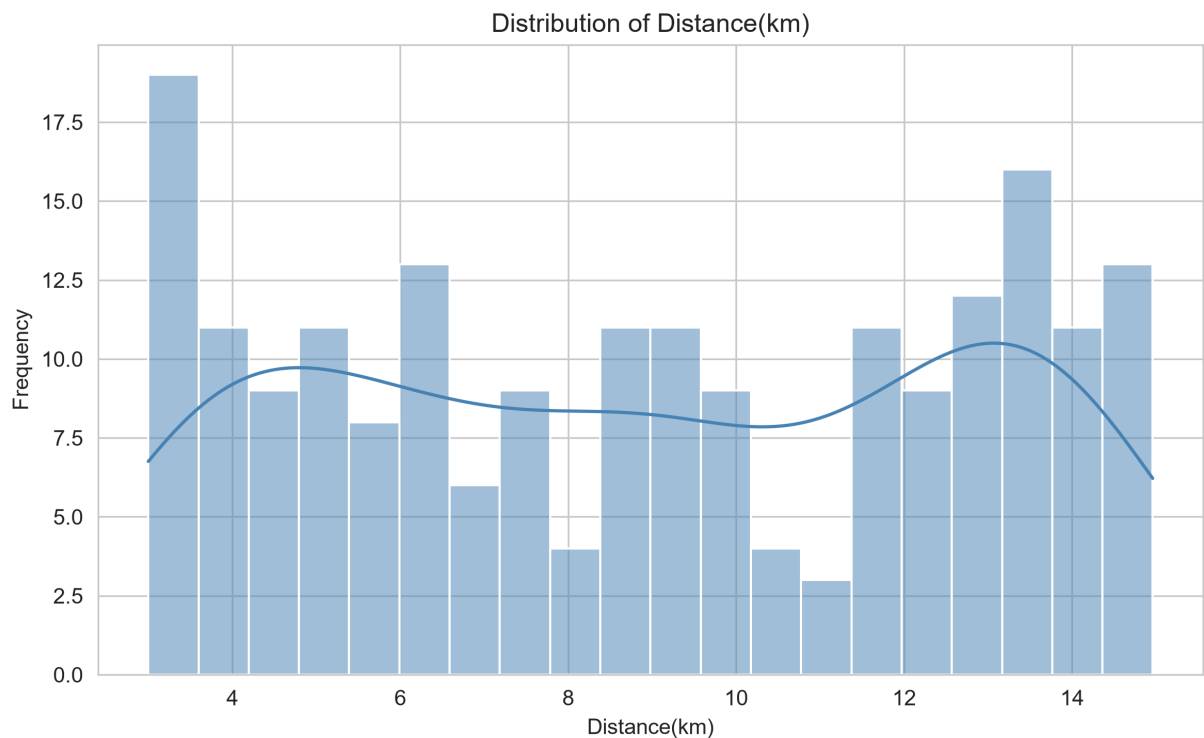


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Average_Heart_Rate.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Average_Heart_Rate.png

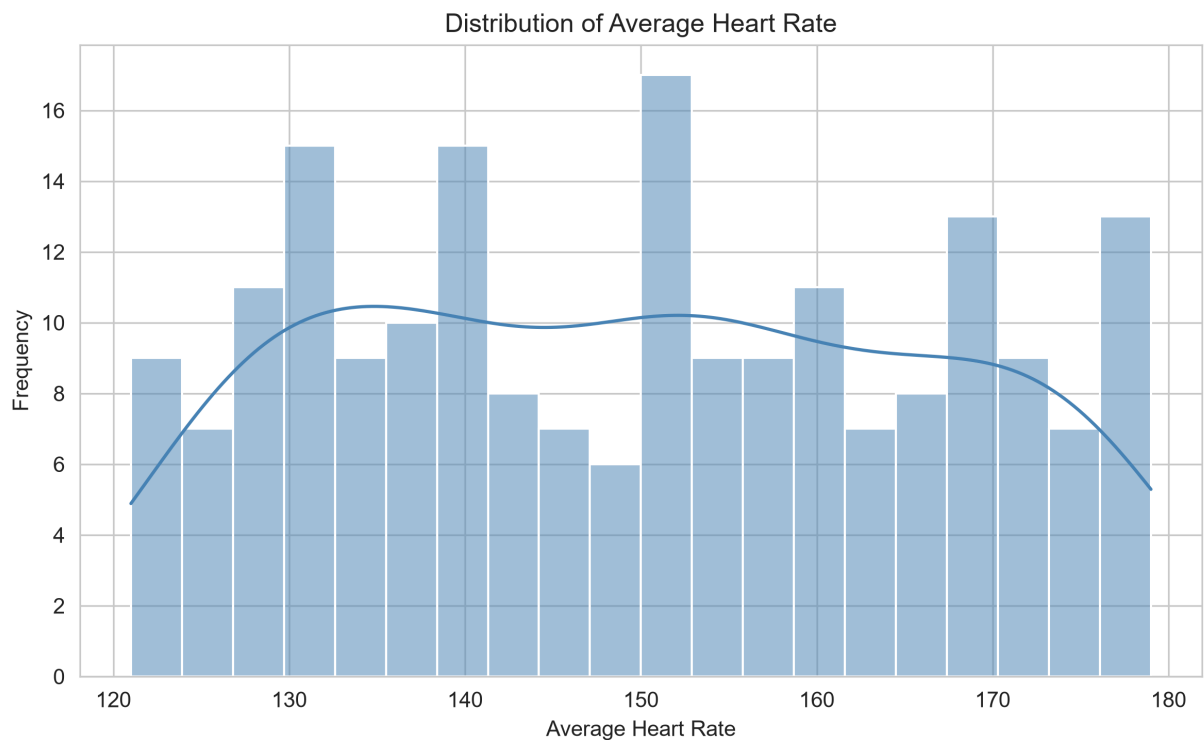
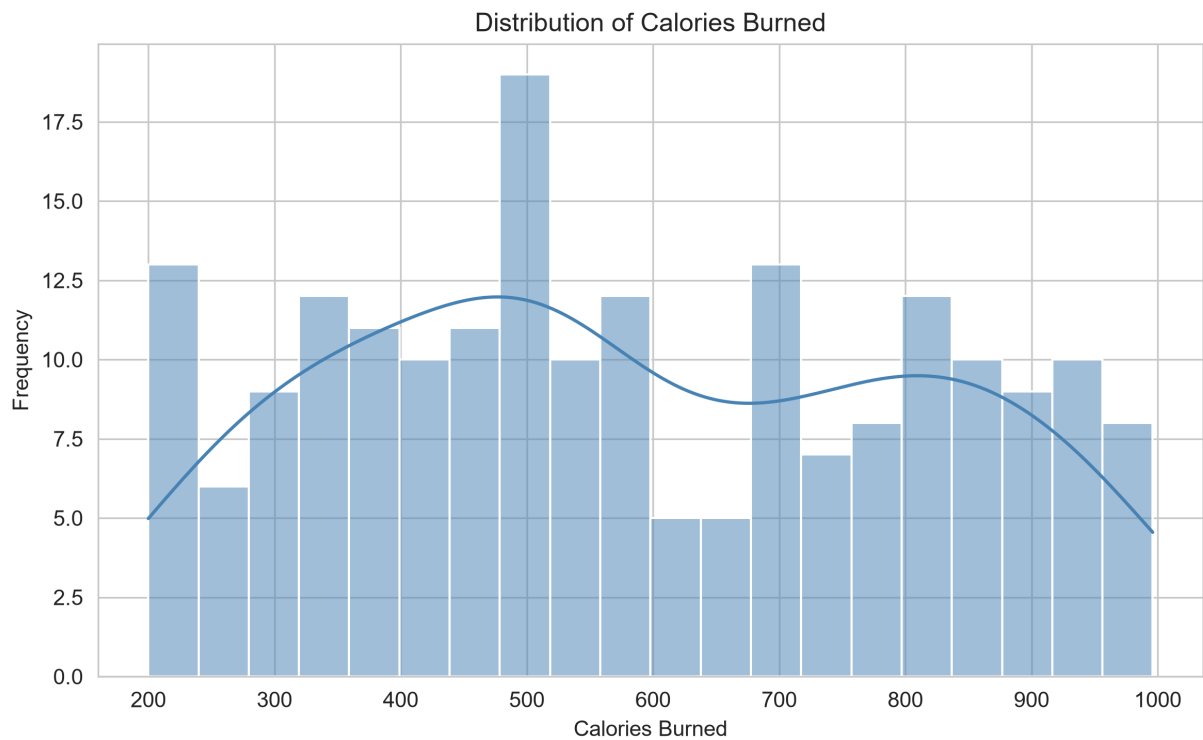


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\histogram_Calories_Burned.png



Numerical Feature Boxplots — `plot_boxplot()`

```
In [66]: for feature in numerical_features:  
         call_and_show(plot_boxplot, df, feature)
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Age.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Age.png

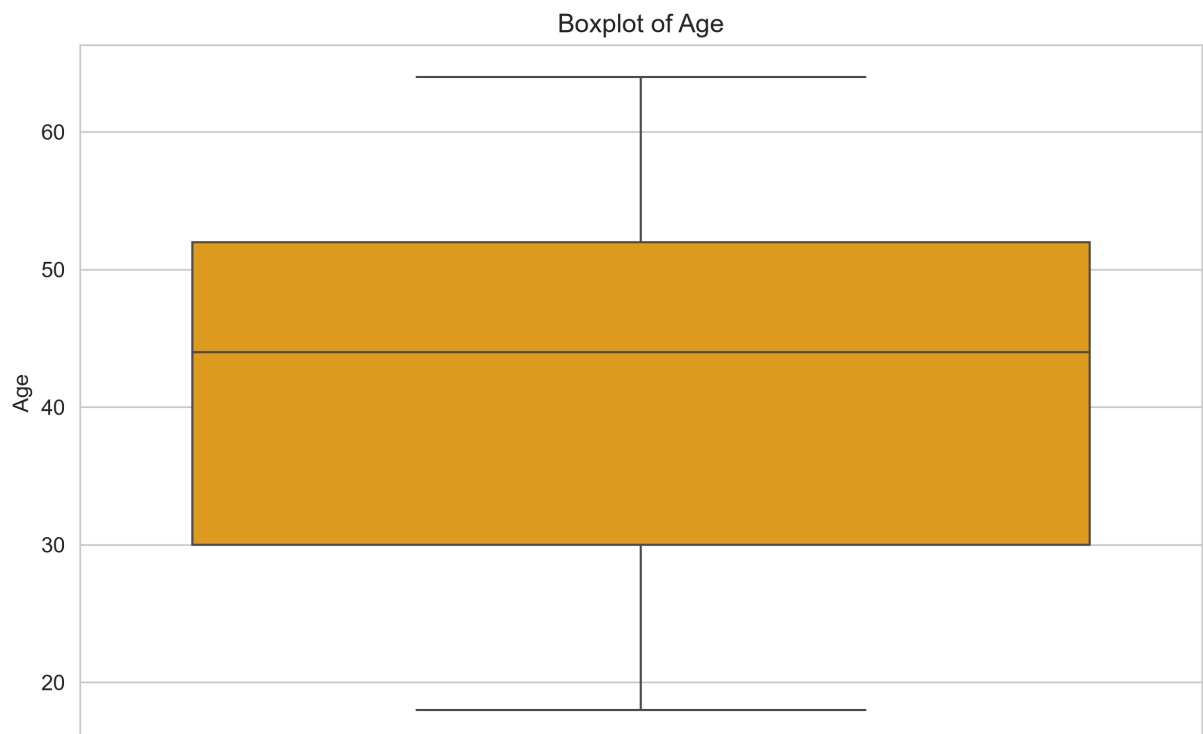


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Height(cm).png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Height(cm).png



Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Weight(kg).png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Weight(kg).png

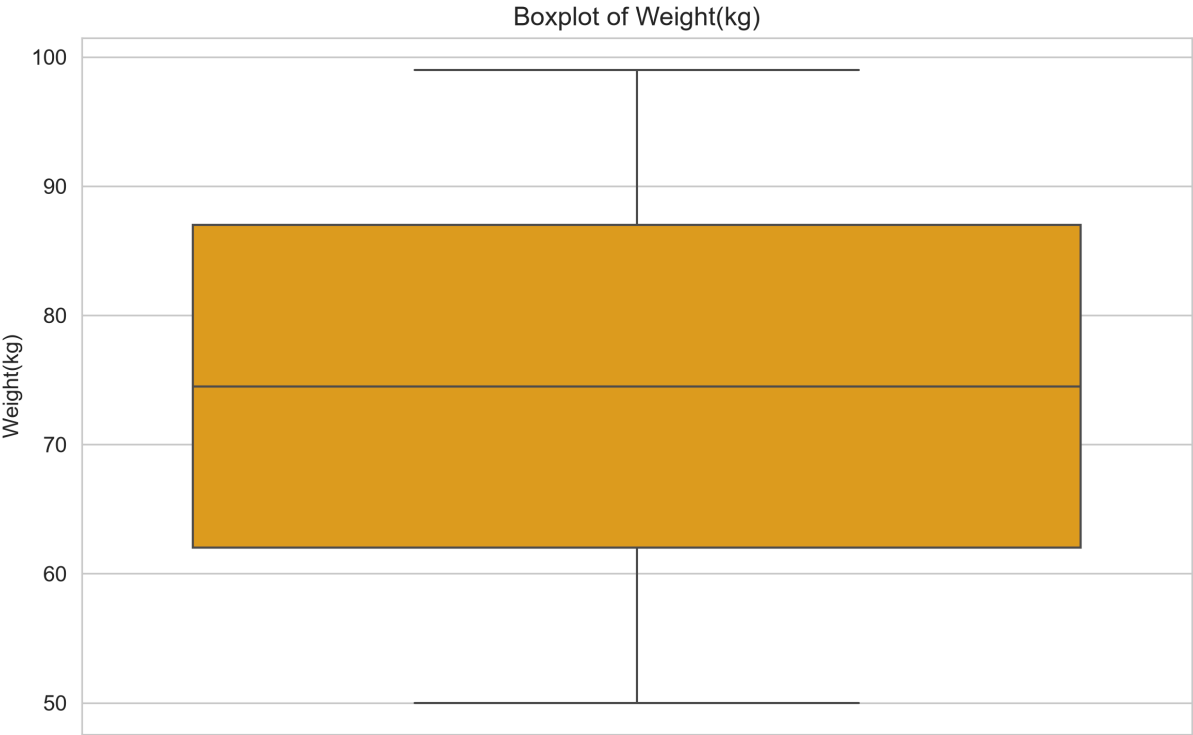


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_BMI.png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_BMI.png

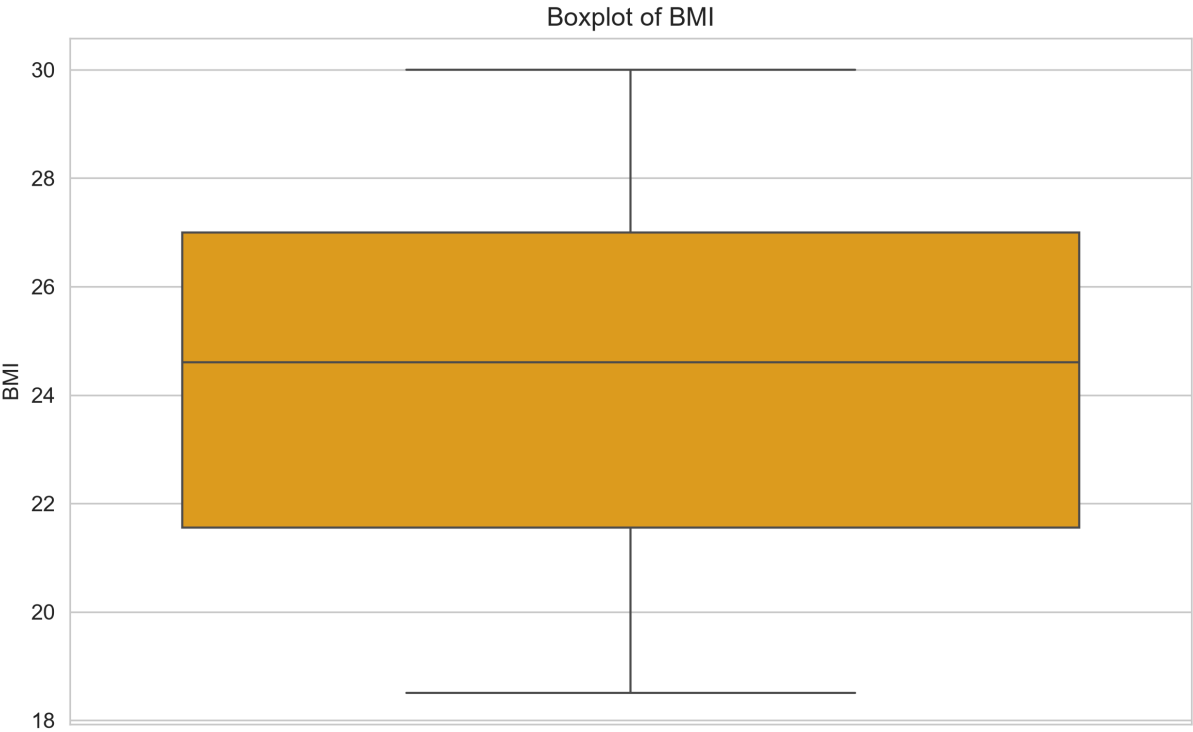


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Running_Time(min).png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Running_Time(min).png

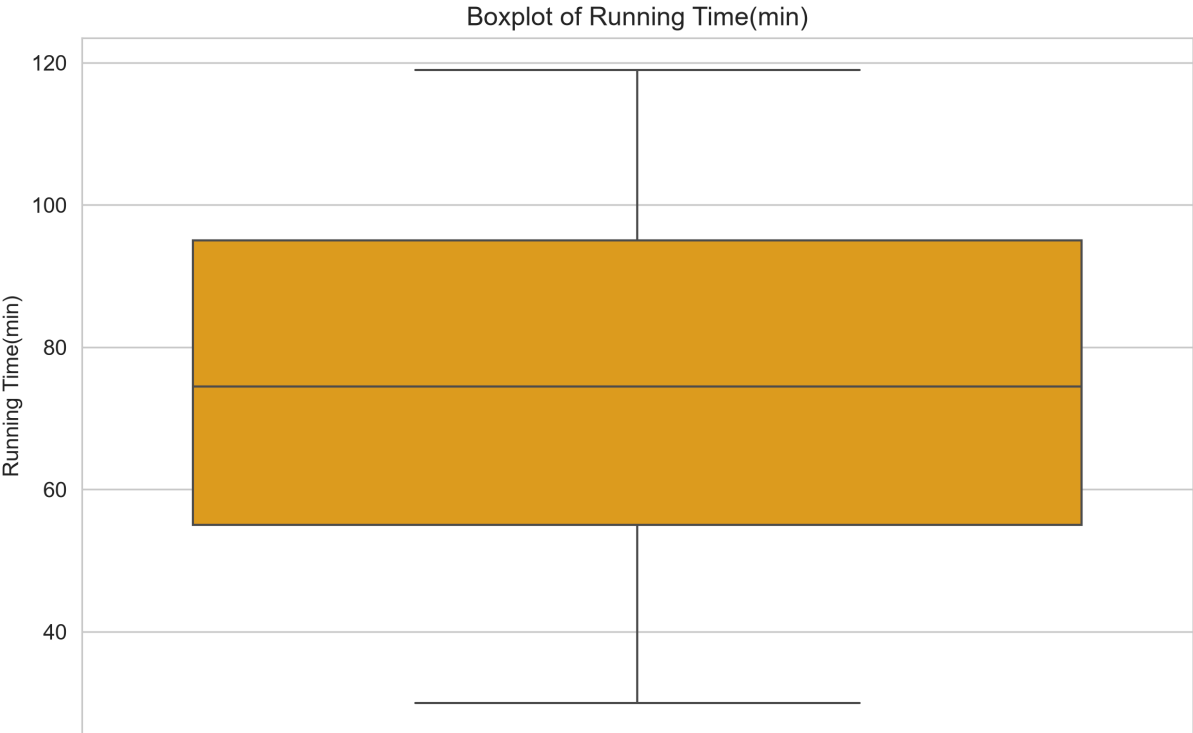


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Running_Speed(km_h).png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Running_Speed(km_h).png

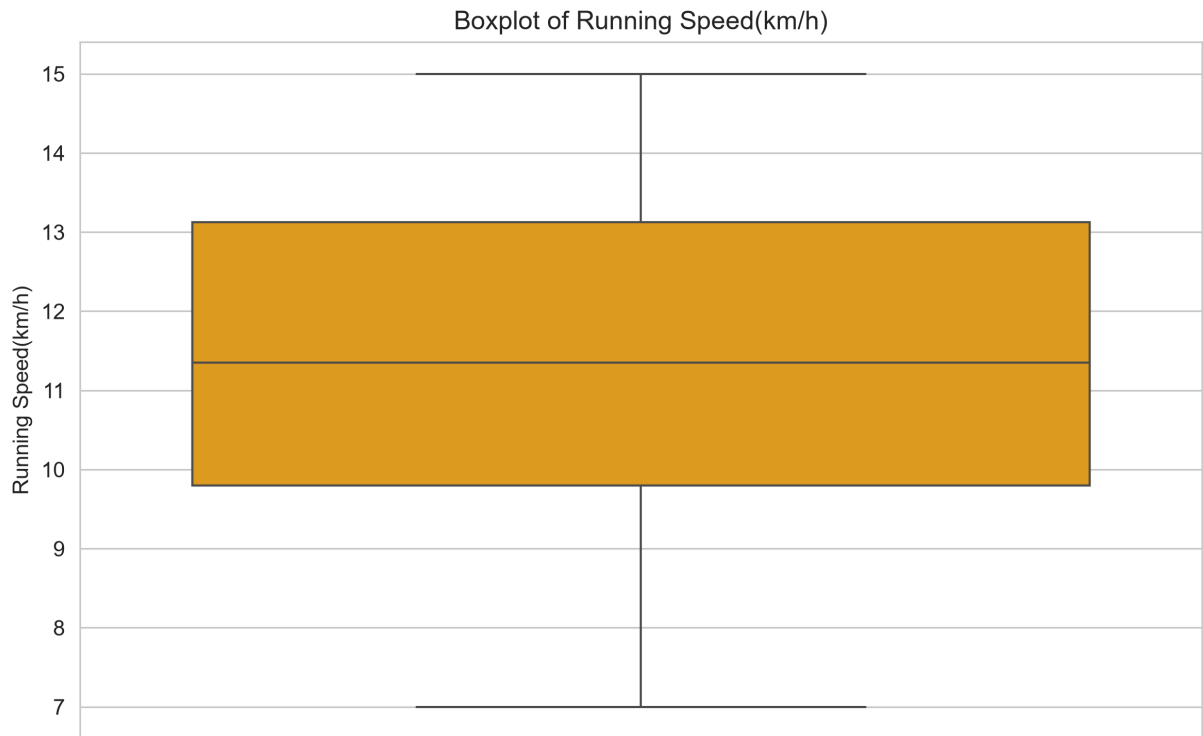


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Distance(km).png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Distance(km).png

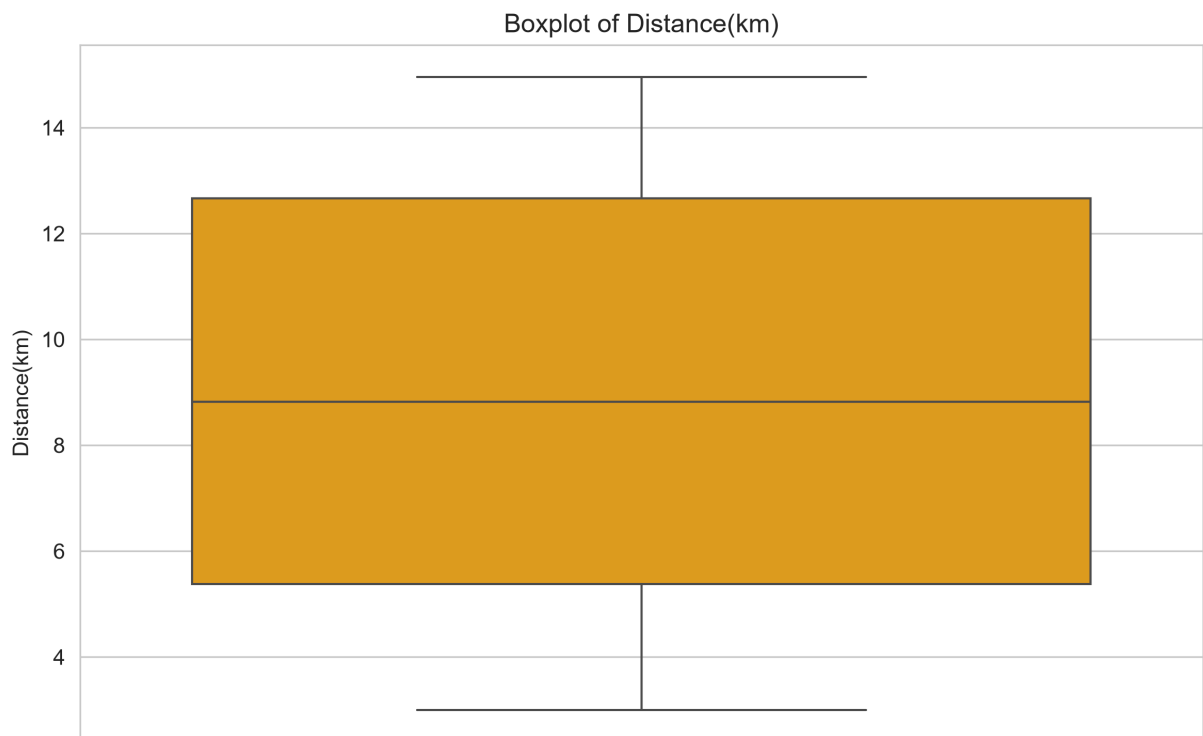


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Average_Heart_Rate.png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Average_Heart_Rate.png

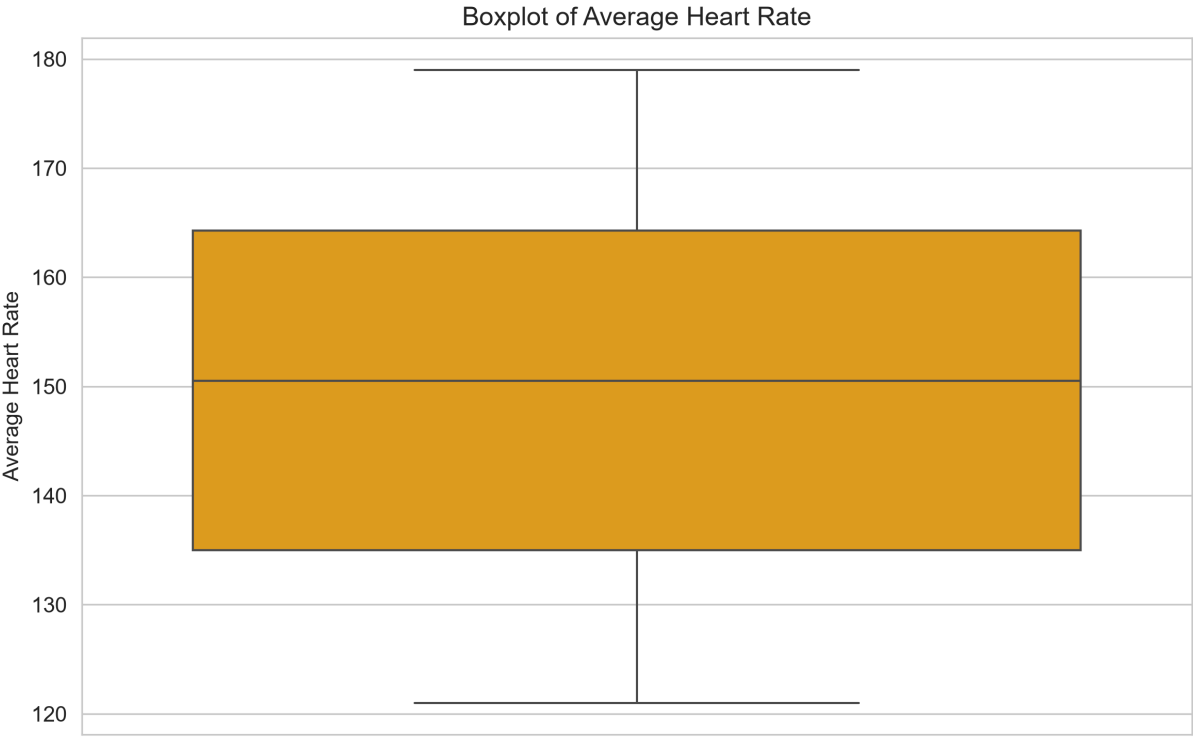


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Calories_Burned.png
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\boxplot_Calories_Burned.png

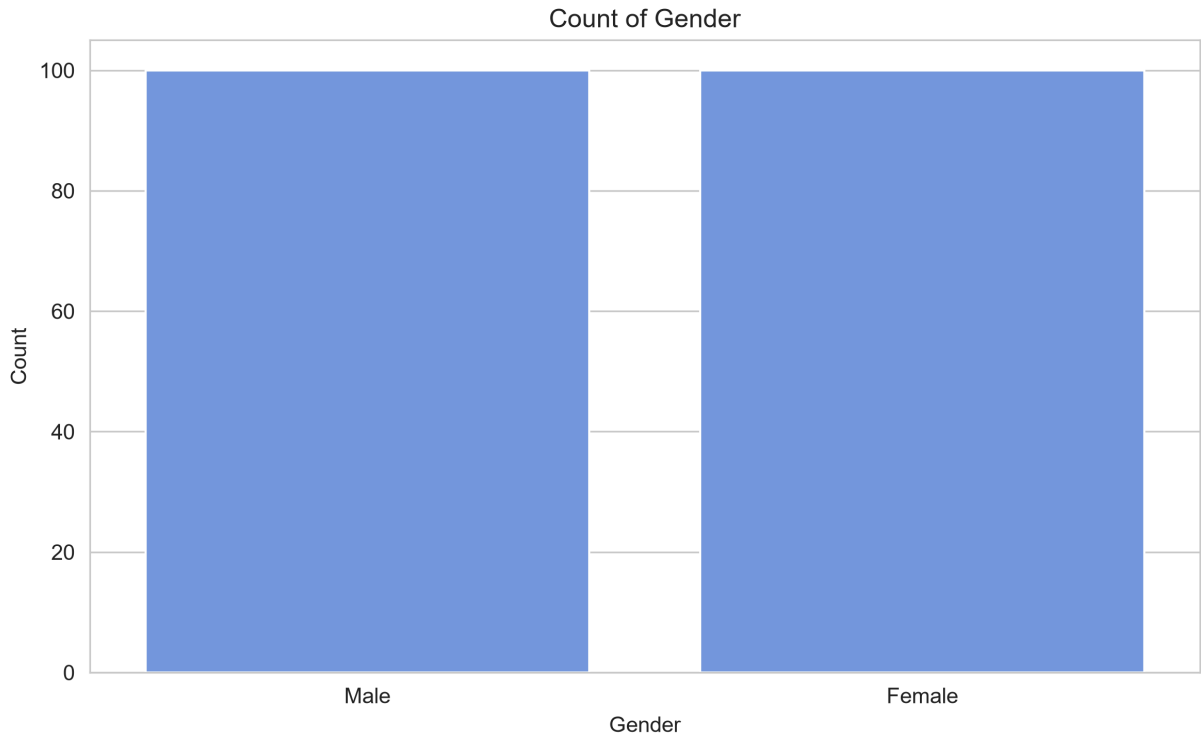


Gender Distribution — `plot_countplot()`

```
In [67]: call_and_show(plot_countplot, df, "Gender")  
print(df["Gender"].value_counts())
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\countplot_Gender.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\countplot_Gender.png



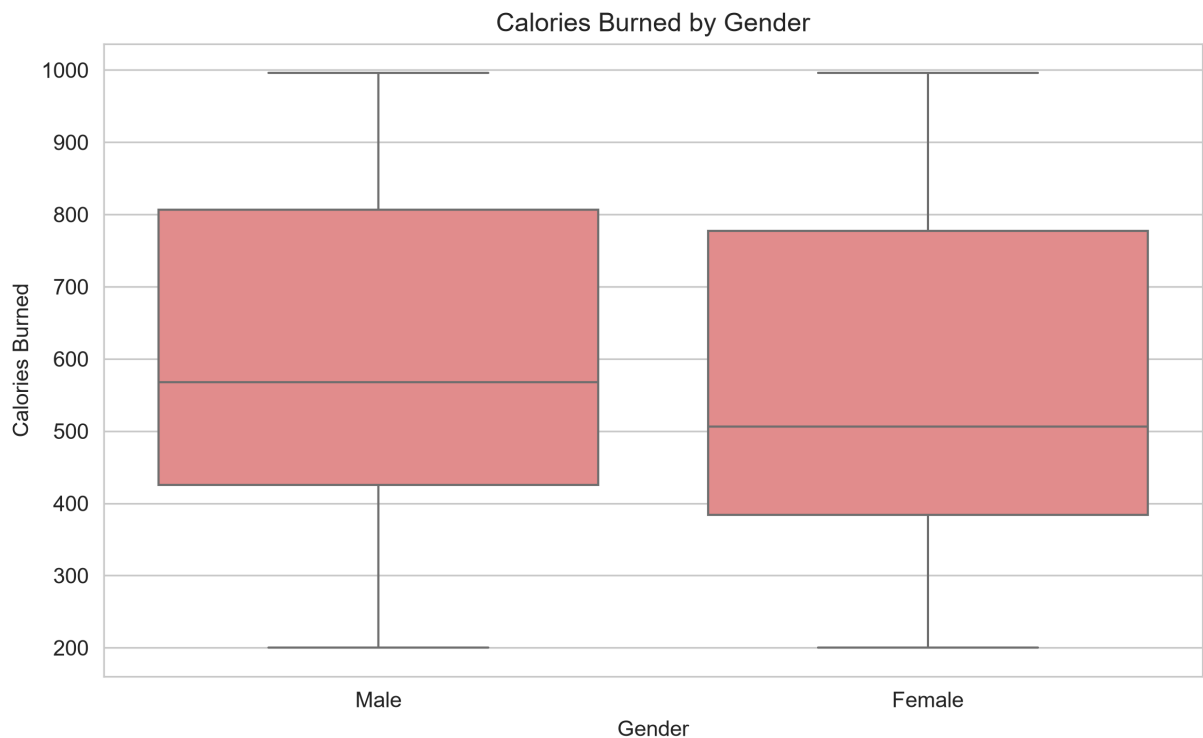
```
Gender  
Male      100  
Female    100  
Name: count, dtype: int64
```

Calories Burned by Gender — `plot_calories_by_gender()` / `plot_violin()`

```
In [68]: call_and_show(plot_calories_by_gender, df)
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\calories_by_gender.png

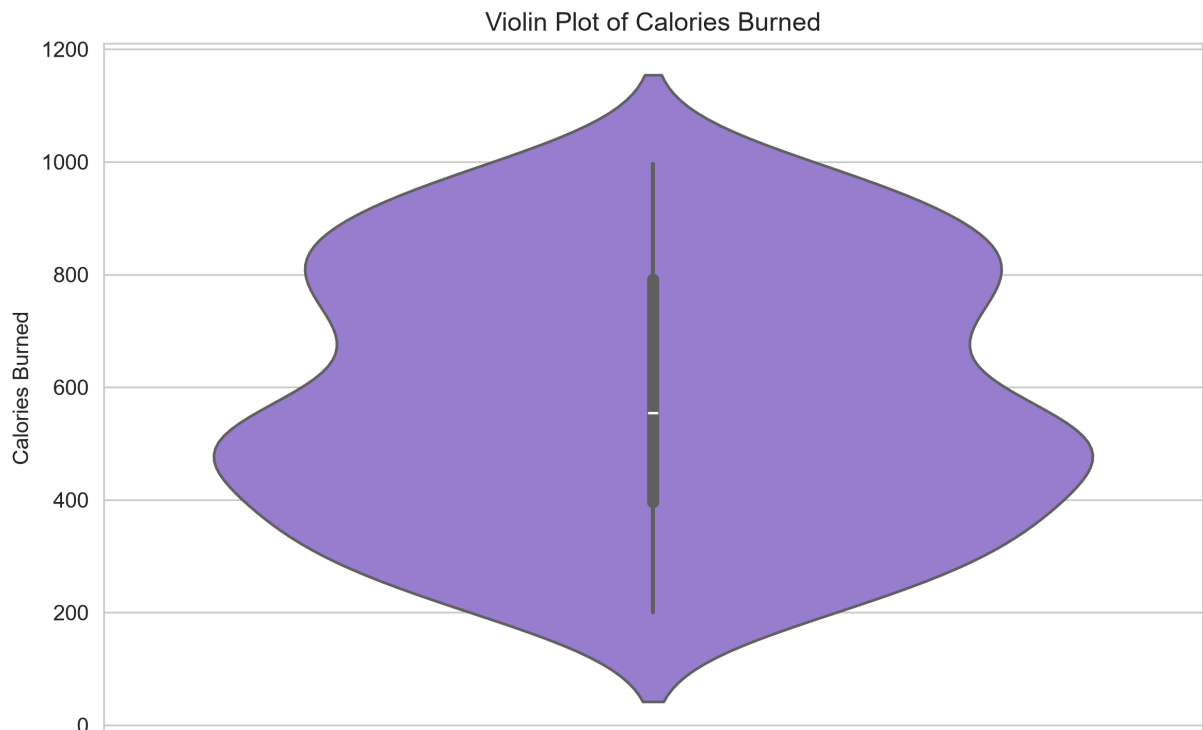
Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\calories_by_gender.png



```
In [69]: call_and_show(plot_violin, df, "Calories Burned")
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\violin_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\violin_Calories_Burned.png

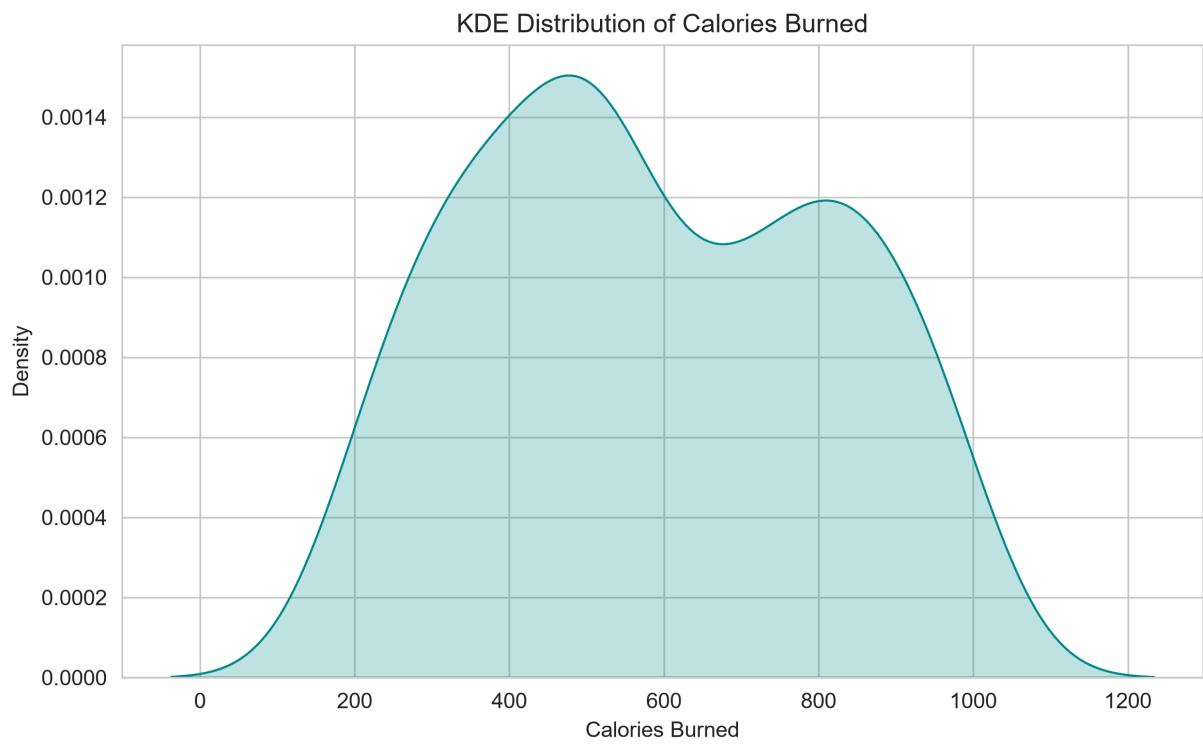


Calories Burned — KDE — `plot_kde()`

```
In [70]: call_and_show(plot_kde, df, "Calories Burned")
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\kde_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\kde_Calories_Burned.png



Relationships With the Target — `plot_calories_vs_*`()

```
In [71]: call_and_show(plot_calories_vs_time, df)
call_and_show(plot_calories_vs_distance, df)
call_and_show(plot_calories_vs_heart_rate, df)
call_and_show(plot_calories_vs_weight, df)
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Running_Time(min)_vs_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Running_Time(min)_vs_Calories_Burned.png

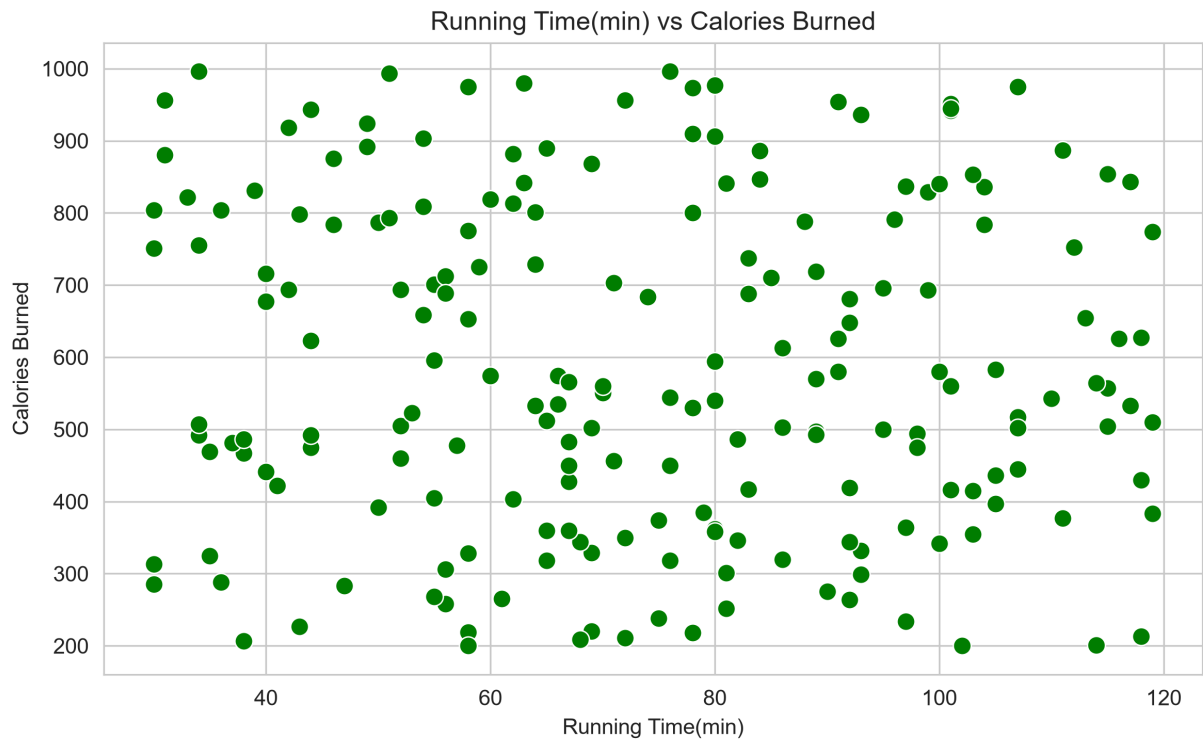


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Distance(km)_vs_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Distance(km)_vs_Calories_Burned.png



Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Average_Heart_Rate_vs_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Average_Heart_Rate_vs_Calories_Burned.png

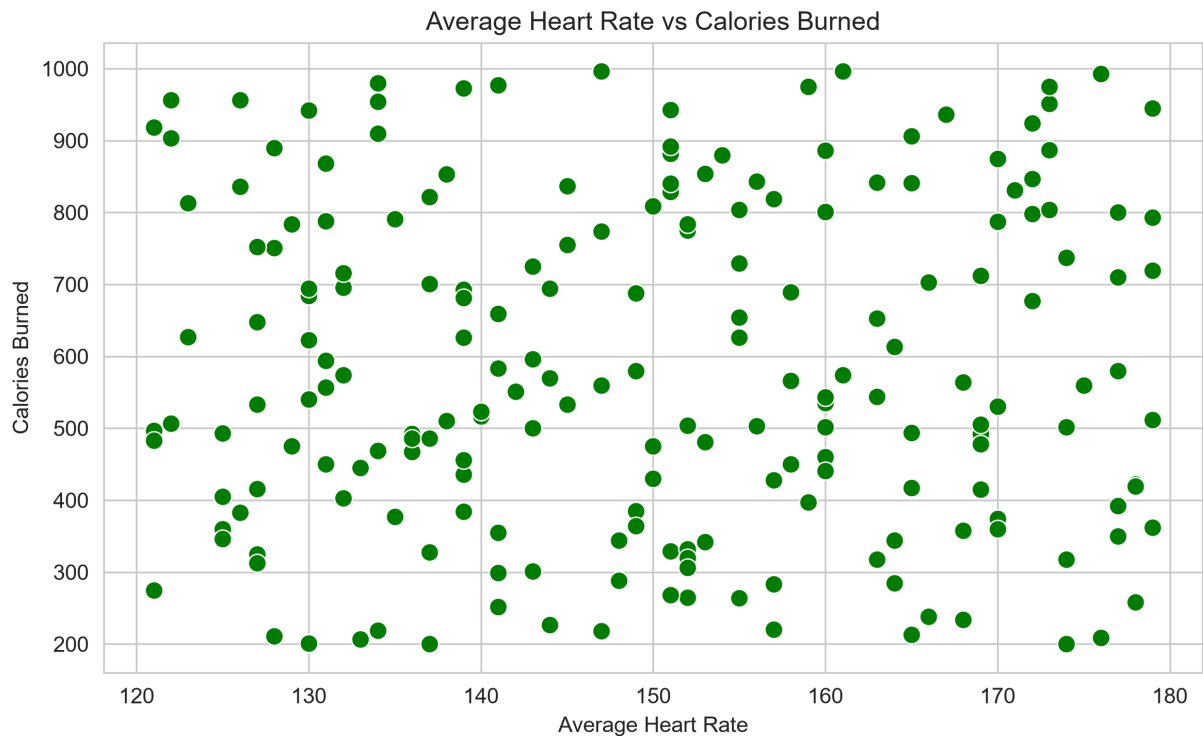


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_weight(kg)_vs_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_weight(kg)_vs_Calories_Burned.png

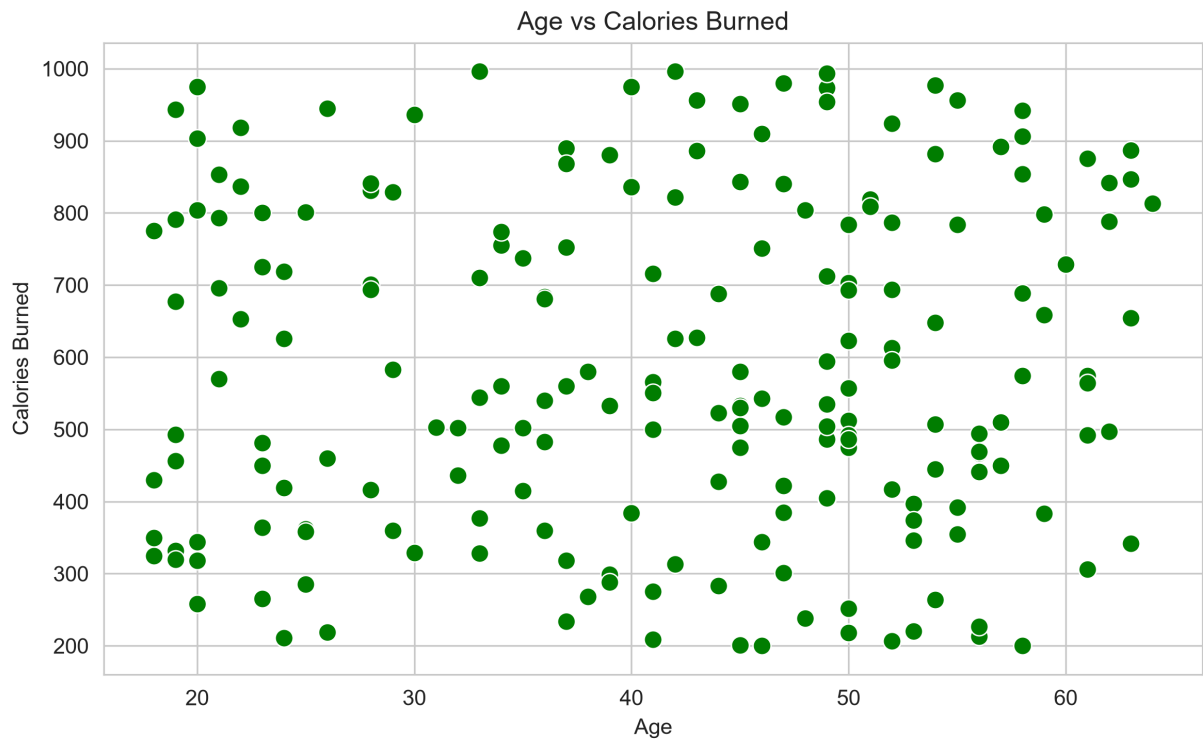


Calories vs Age — `plot_scatter()`

```
In [72]: call_and_show(plot_scatter, df, "Age", "Calories Burned")
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Age_vs_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Age_vs_Calories_Burned.png



Feature-to-Feature Relationships — `plot_scatter()`

```
In [73]: call_and_show(plot_scatter, df, "Running Time(min)", "Distance(km)")  
         call_and_show(plot_scatter, df, "Running Speed(km/h)", "Distance(km)")
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Running_Time(min)_vs_Distance(km).png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Running_Time(min)_vs_Distance(km).png

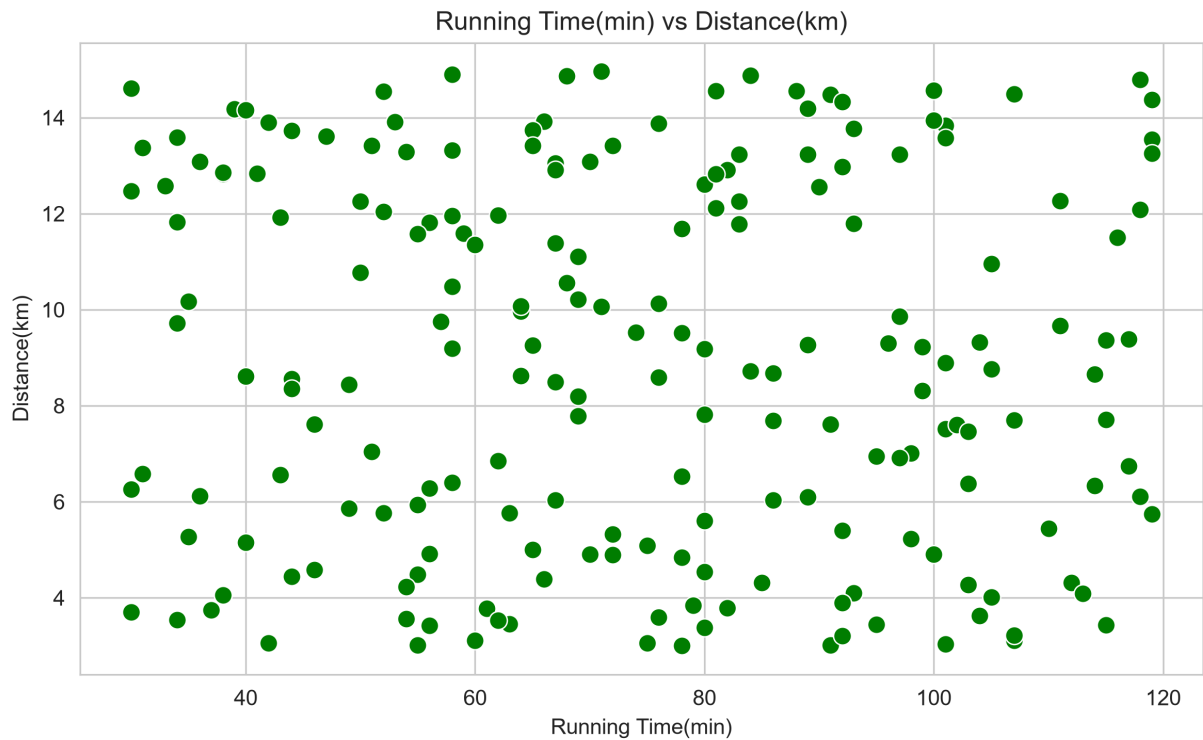
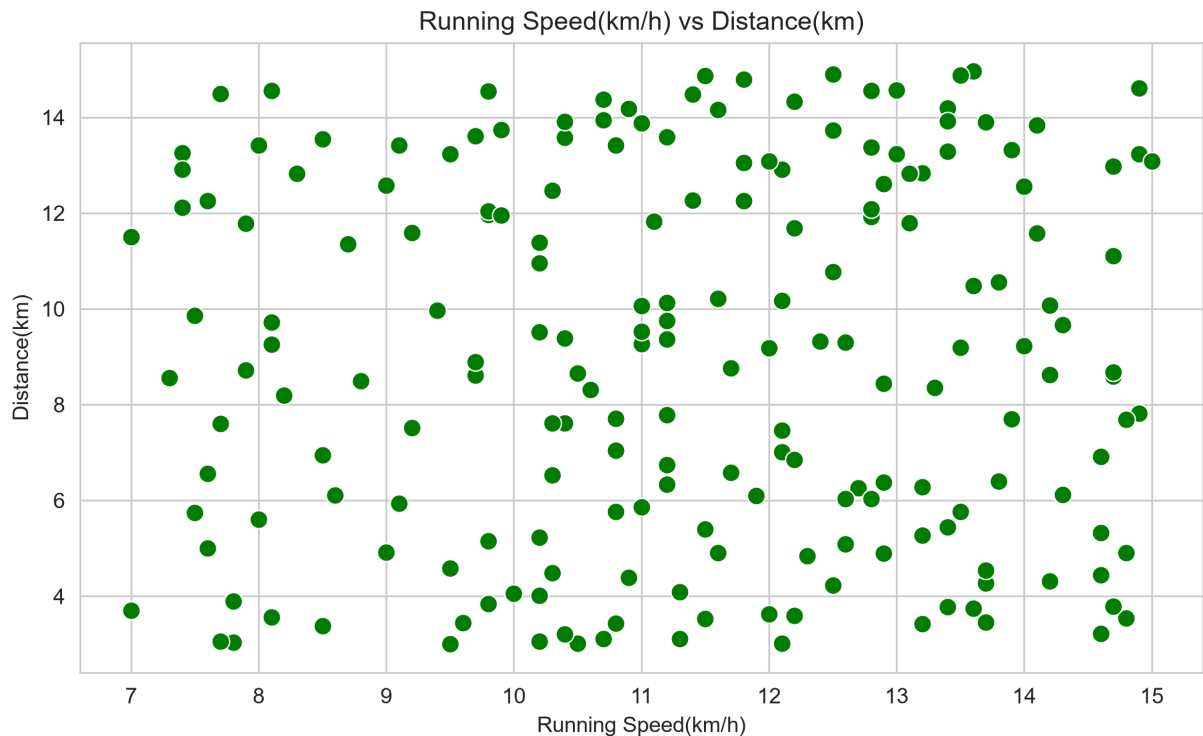


Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Running_Speed(km_h)_vs_Distance(km).png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_Running_Speed(km_h)_vs_Distance(km).png

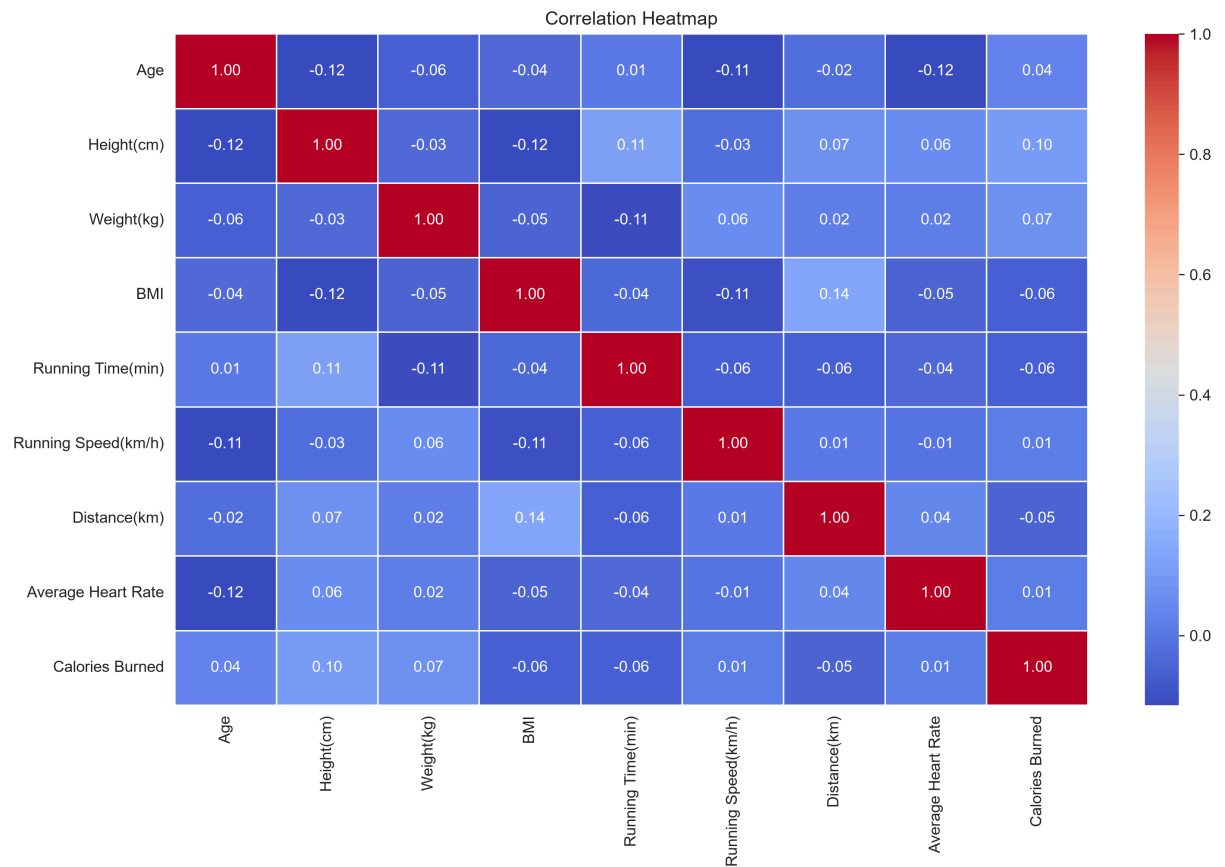


Correlation Structure — `plot_correlation_heatmap()` / `plot_target_correlations()`

```
In [74]: call_and_show(plot_correlation_heatmap, df)
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\correlation_heatmap.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\correlation_heatmap.png



```
In [75]: call_and_show(plot_target_correlations, df, target_column="Calories Burned")

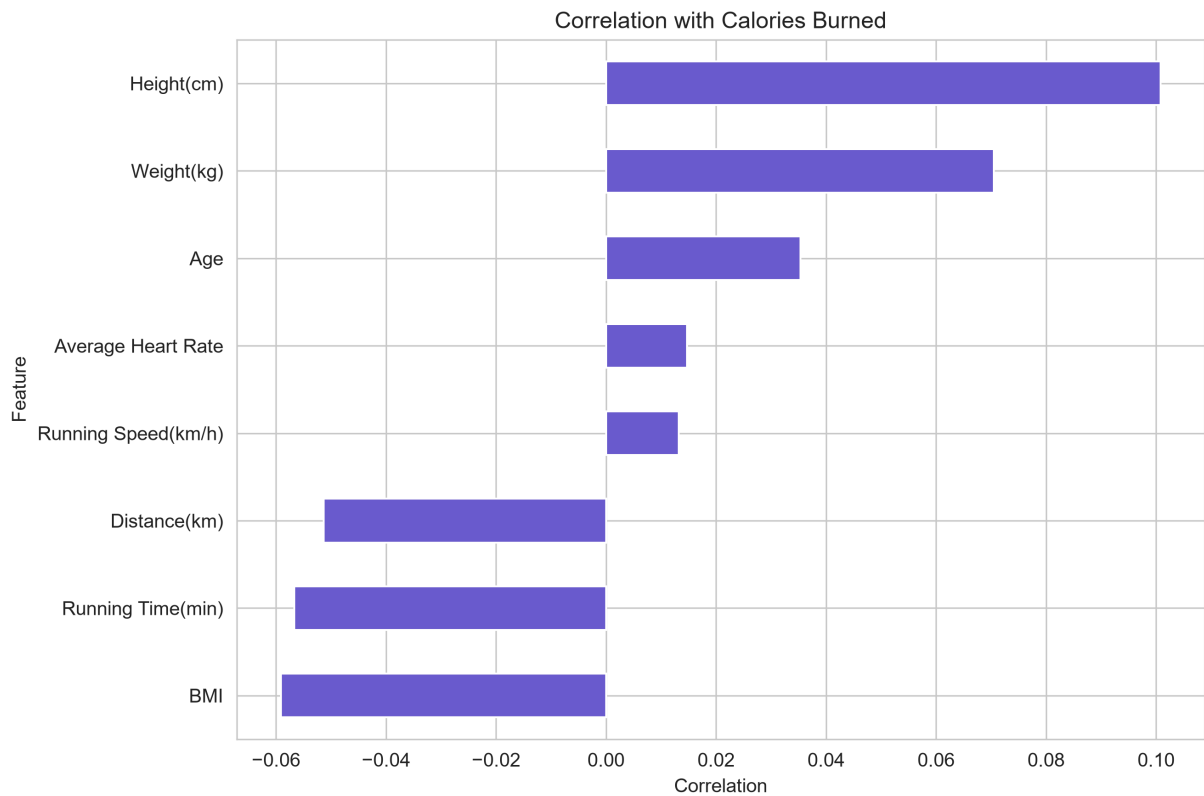
target_correlations = (
    correlation_matrix["Calories Burned"]
    .drop("Calories Burned")
)

correlation_table = pd.DataFrame({
    "Feature": target_correlations.index,
    "Correlation": target_correlations.values,
    "Absolute Correlation": target_correlations.abs().values,
}).sort_values("Absolute Correlation", ascending=False).reset_index(drop=True)

correlation_table
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\target_correlations.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\target_correlations.png



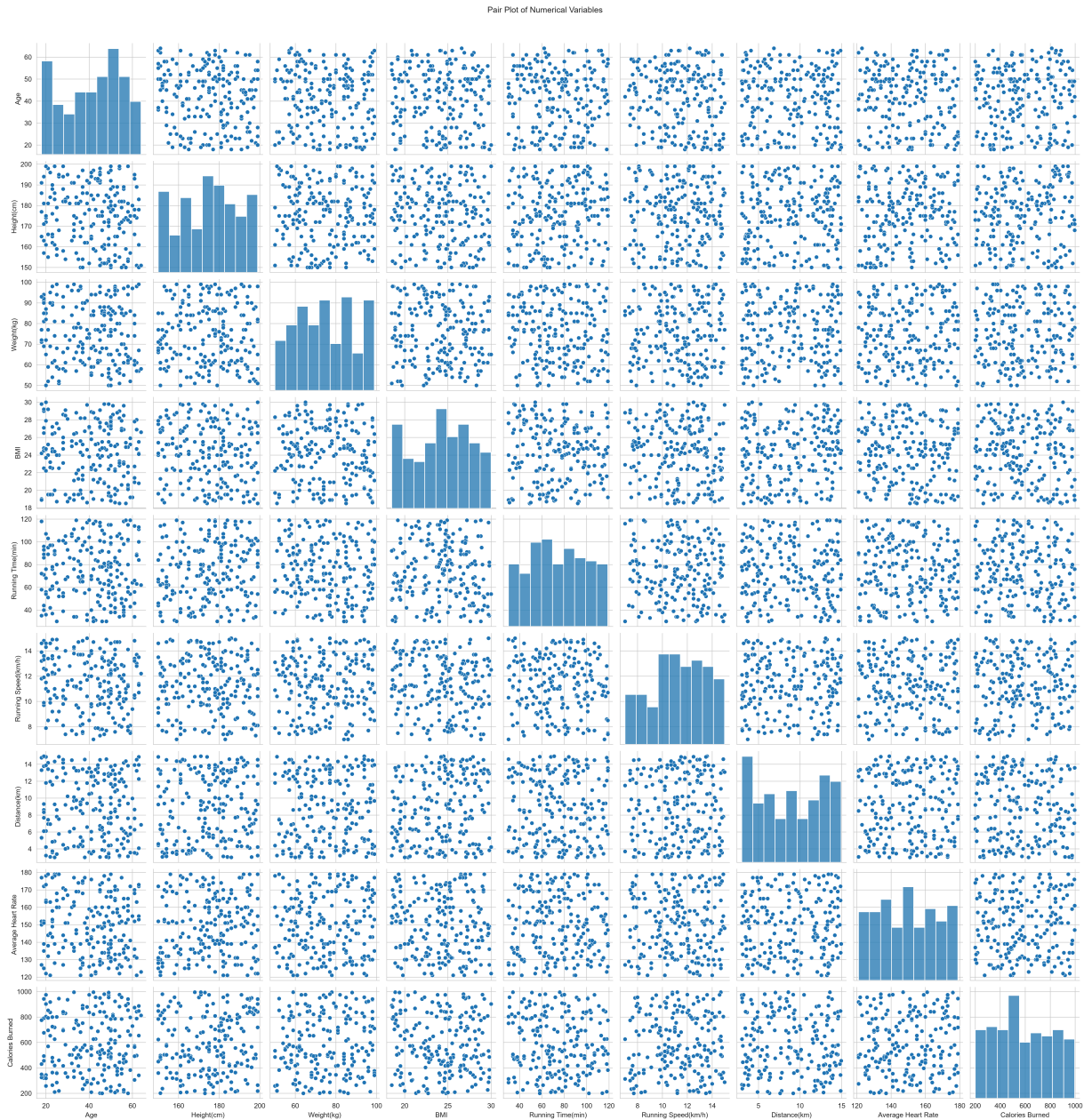
Out[75]:

	Feature	Correlation	Absolute Correlation
0	Height(cm)	0.100808	0.100808
1	Weight(kg)	0.070512	0.070512
2	BMI	-0.059128	0.059128
3	Running Time(min)	-0.056699	0.056699
4	Distance(km)	-0.051363	0.051363
5	Age	0.035348	0.035348
6	Average Heart Rate	0.014701	0.014701
7	Running Speed(km/h)	0.013207	0.013207

Pair Plot — `plot_pairplot()`

In [76]: `call_and_show(plot_pairplot, df)`

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\pairplot.png
 Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\pairplot.png

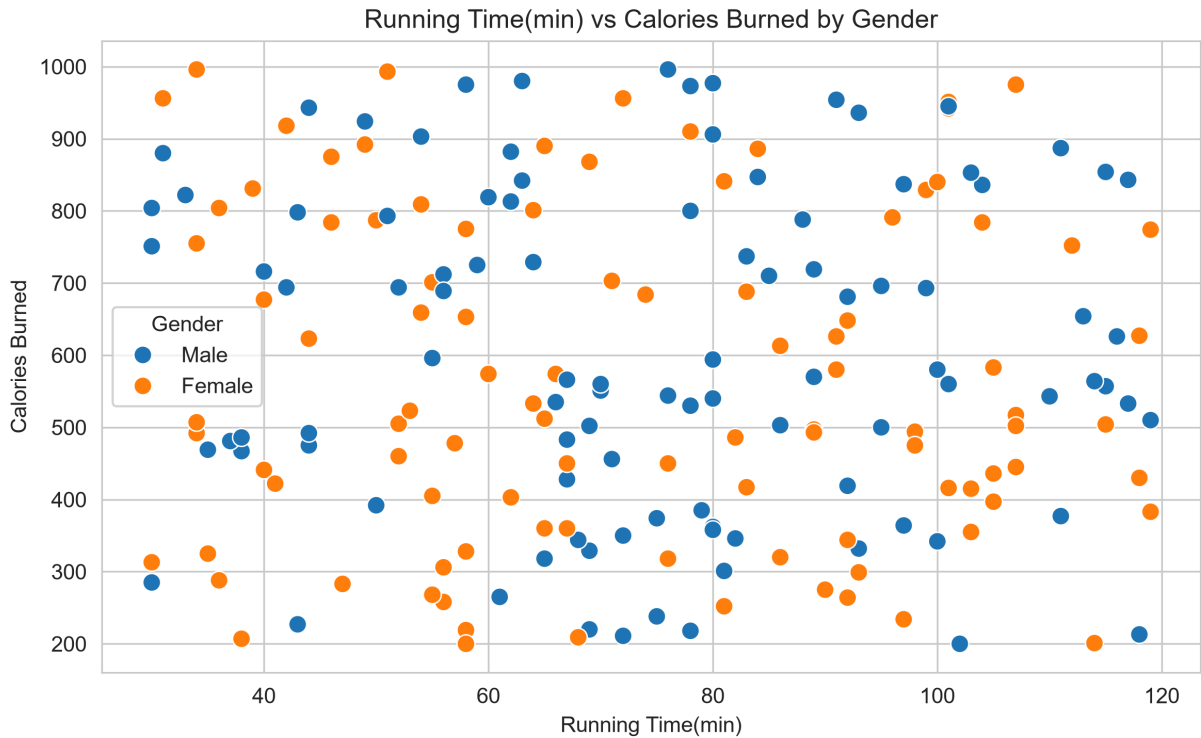


Running Time vs Calories Burned, by Gender — plot_scatter_by_gender()

```
In [77]: call_and_show(plot_scatter_by_gender, df, "Running Time(min)", "Calories Burned")
```

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_gender_Running_Time(min)_vs_Calories_Burned.png

Figure saved: C:\Users\ayman\OneDrive\Desktop\calories-prediction\reports\figures\scatter_gender_Running_Time(min)_vs_Calories_Burned.png



Summary

English

- **Loading:** `src/data_loader.py` resolves paths from the project root and loads both the raw and processed datasets.
- **Analysis:** `src/data_analysis.py` produced the full before/after report — basic info, missing values, duplicates, statistics, and IQR-based outlier counts.
- **Cleaning:** `src/data_cleaning.py`'s `run_cleaning_pipeline()` handled standardization, type conversion, missing values, duplicates, and invalid/impossible values in one call, and saved `data/processed/dataset_clean.csv`.
- **Visualization:** every chart in this notebook came from `src/visualization.py`; none of the plotting logic is duplicated here.
- **Next step:** the correlation table above (sorted by absolute correlation with `Calories Burned`) is the natural starting point for feature selection in the modeling stage.

العربية

- **التحميل:** `src/data_loader.py` بتحديد المسارات انطلاقًا من جذر المشروع، وتحمل البيانات الخام والمعالجة.
- **التحليل:** `src/data_analysis.py` التقرير الكامل قبل/بعد — المعلومات الأساسية، IQR القيم المفقودة، التكرارات، الإحصاءات، وعدد القيم المتطرفة بطريقة.
- **التنظيف:** `run_cleaning_pipeline()` من `src/data_cleaning.py` التوحيد وتحويل الأنواع والقيم المفقودة والتكرارات والقيم غير الصالحة/المستحيلة في استدعاء واحد، وحفظت `data/processed/dataset_clean.csv`.

- لا يوجد `src/visualization.py` **التصور البصري**: جميع الرسومات في هذا الدفتر مصدرها أي تكرار لمنطق الرسم هنا.
- `Calories` مرتب حسب القيمة المطلقة للارتباط مع **الخطوة التالية**: جدول الارتباط أعلاه. هو نقطة الانطلاق الطبيعية لاختيار الخصائص في مرحلة النمذجة (`Burned`)

Generated Figures

English

A final listing of every PNG currently saved in `reports/figures/` , to confirm everything this notebook produced (via `src/visualization.py`) is actually on disk.

العربية

للتأكد من أن كل ما أنتجه هذا ، `reports/figures/` محفوظ حاليًا في PNG قائمة نهائية بكل ملف موجود فعليًا على القرص (`src/visualization.py` عبر) الدفتر

```
In [78]: figure_files = sorted(FIGURES_DIR.glob("*.png"))

figures_summary = pd.DataFrame({
    "Filename": [f.name for f in figure_files],
    "Path": [str(f) for f in figure_files],
    "File Size (KB)": [round(f.stat().st_size / 1024, 2) for f in figure_files],
})

print(f"Total figures found: {len(figures_summary)}")
figures_summary
```

Total figures found: 83

Out[78]:

	Filename	Path	File Size (KB)
0	bar_chart_Gender.png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	46.57
1	boxplot_Age.png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	37.70
2	boxplot_Average_Heart_Rate.png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	53.97
3	boxplot_BMI.png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	37.24
4	boxplot_Calories_Burned.png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	53.46
...	...	...	...
78	violin_Distance(km).png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	117.07
79	violin_Height(cm).png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	116.54
80	violin_Running_Speed(km_h).png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	115.58
81	violin_Running_Time(min).png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	116.10
82	violin_Weight(kg).png	C:\Users\ayman\OneDrive\Desktop\calories-predi...	112.03

83 rows × 3 columns